Check for updates

AGU100 ADVANCING EARTH AND SPACE SCIENCE

Geophysical Research Letters

RESEARCH LETTER
10.1029/2019GL085473

Direct Paleomagnetic Constraint on the Closure of Paleo-Tethys and Its Implications for Linking the Tibetan and Southeast Asian Blocks

Yonggang Yan[1,2], Qian Zhao[3], Yipeng Zhang[1,2], Baochun Huang[3], Wenjun Zheng[1,2], and Peizhen Zhang[1,2]

[1]Guangdong Provincial Key Laboratory of Geodynamics and Geohazards, School of Earth Sciences and Engineering, Sun Yat-sen University, Guangzhou, China, [2]Southern Laboratory of Ocean Science and Engineering, Zhuhai, China, [3]Key Laboratory of Orogenic Belt and Crust Evolution, Ministry of Education, School of Earth and Space Sciences, Peking University, Beijing, China

Key Points:
- The southern Lancangjiang volcanic belt formed at a paleolatitude of ~26ºN during the Carnian Stage of the Late Triassic
- The north branch of the Paleo-Tethys closed no later than the Carnian Stage of the Late Triassic
- A Permo-Triassic geographic linkage between Indochina and N-Qiangtang is confirmed

Supporting Information:
- Supporting Information S1
- Table S1

Correspondence to:
Y. Yan,
yanyongg@mail.sysu.edu.cn

Citation:
Yan, Y., Zhao, Q., Zhang, Y., Huang, B., Zheng, W., & Zhang, P. (2019). Direct paleomagnetic constraint on the closure of Paleo-Tethys and its implications for linking the Tibetan and Southeast Asian Blocks. *Geophysical Research Letters*, *46*, 14,368–14,376. https://doi.org/10.1029/2019GL085473

Received 30 SEP 2019
Accepted 6 DEC 2019
Accepted article online 10 DEC 2019
Published online 18 DEC 2019

Abstract Paleomagnetic studies on the syncollisional magmatic eruptions provide a direct way to understand when and where the Paleo-Tethys closed. We report a well-dated paleomagnetic pole from the southern Lancangjiang volcanic belt in western Yunnan of China. Stable Characteristic Remanent Magnetizations (ChRMs) were isolated from 21 sites at high temperatures following step-wise thermal demagnetization. The data pass fold, reversal, and conglomerate tests and are interpreted to be primary thermal remanent magnetizations. The new paleomagnetic pole (46.1°N/176.2°E, K=44.9, A_{95}=4.8°, N=21) yields a paleolatitude of 26.0 ± 4.8°N at the reference site (23.0°N/100.4°E) in the southern Lancangjiang area. This direct paleo-latitudinal constraint on the Paleo-Tethys suture zone in western Yunnan indicates that the north branch of the Paleo-Tethys Ocean closed no later than the Carnian Age of Late Triassic, and confirms a geographic linkage between the North Qiangtang and Indochina Blocks at least during the Early Permian to Late Triassic.

Plain Language Summary The Asian continent is formed by the amalgamation of continental blocks such as the Tarim, North China, South China, Indochina, and Qiangtang. However, questions remained to answer: When did the ocean between them close? How did they drift from the south hemisphere to the north? In this study, we present a paleomagnetic constraint on the Paleo-Tethys suture zone to quantify when and where the Paleo-Tethys Ocean closed. The Paleo-Tethys Ocean closed no later than the early Late Triassic at ~26°N. The Indochina and North Qiangtang drifted together from the south hemisphere to the north during the Permian-Triassic.

1. Introduction

Reconstructions of Pangea during late Paleozoic to Mesozoic times indicate that a series of continental blocks rifted from the northern margin of Gondwana, drifted through the Tethys Ocean and eventually accreted into Eurasia (Metcalfe, 2013; Torsvik et al., 2012). Paleomagnetically quantifying the rift-drift history of the Gondwana-derived blocks can provide critical information for understanding their paleogeographic linkages, which are essential to reconstruct the evolution of the Tethys Ocean. Though geologic and paleomagnetic data have also been obtained from the Alpine (Van Der Voo, 2005), Iranian (Besse et al., 1998; Muttoni et al., 2009), Pakistan (Haag & Heller, 1991), and Pamirs (Xiao et al., 2005) areas, our understanding on the evolution of Paleo-Tethys has been largely dominated by evidences for the early formation of Tibetan Plateau and Southeast Asia peninsula. However, the nature of the linkage between Tibetan and Southeast Asian Blocks varies between different models.

In the Tibetan area, the Qiangtang Block is generally considered to be sutured by the Jinshajiang suture to the north and Bangong-Nujiang suture to the south (Royden et al., 2008; Yin & Harrison, 2000), though it has also been suggested to subdivide into the North Qiangtang and South Qiangtang along the Longmu Co-Shuanghu suture (Figure 1a; Metcalfe, 2013; Xu et al., 2015; Li et al., 2016). The linkage of the Qiangtang Block with the Southeast Asia Blocks can be categorized into two types of models: First, the Qiangtang Block, as a whole, extends southeastward and connects with the Simao-Indochina (Gehrels et al., 2011) or Baoshan-Sibumasu Block (van Hinsbergen et al., 2011). Second, the North Qiangtang-Qamdo-Simao-Indochina is an entire terrane, which is bounded by the Jinshajiang-Ailaoshan-Song Ma suture zone to the northeast against the

©2019. American Geophysical Union. All Rights Reserved.

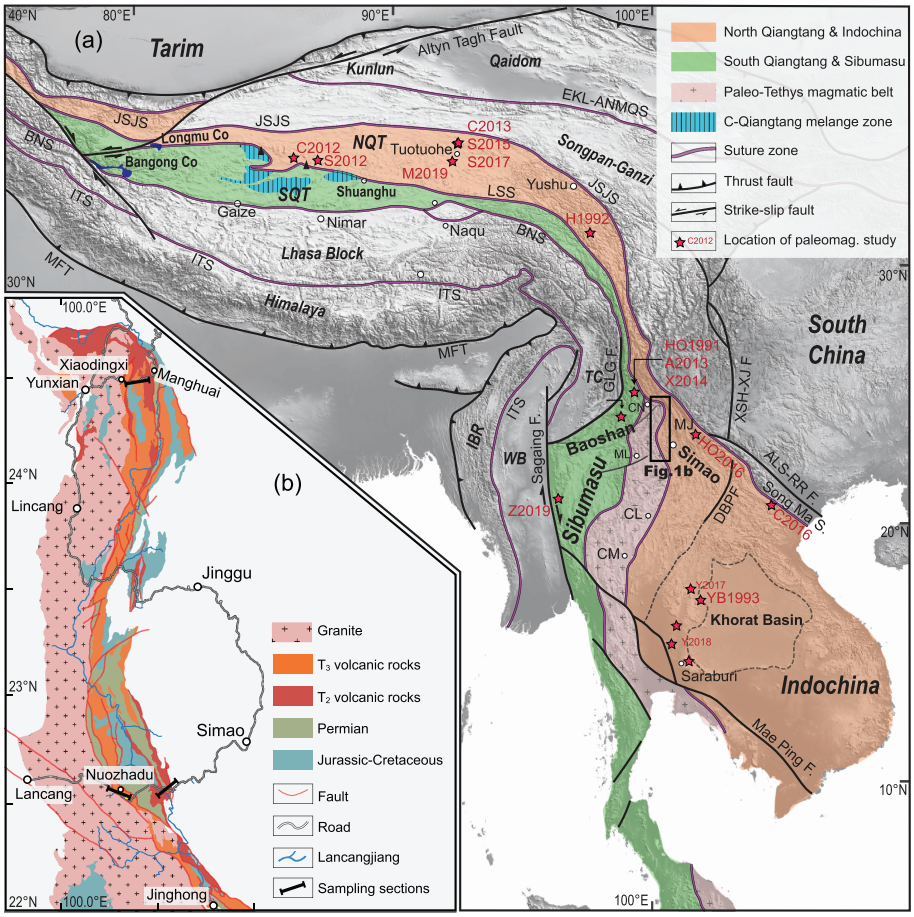


Figure 1. (a) Schematic tectonic map showing the Paleo-Tethys system in the Tibetan and Southeast Asia areas (revised from Ridd et al., 2011; Zhu et al., 2013; Xu et al., 2015). Abbreviation: SQT (South Qiangtang), NQT (North Qiangtang), TC (Tengchong), WB (West Burma), IBR (Indo-Burma Range), EKL-ANMQS (East Kunlun-A'nyemaqen Suture), JSJS (Jinshajiang Suture), LSS (Longmu Co-Shuanghu Suture), BNS (Bangong Co-Nujiang Suture), ITS (Indus-Tsangbu Suture), MFT (Main Frontal Thrust), XSH-XJ F. (Xiaoshuihe-Xiaojiang Fault), ALS-RR F. (Ailaoshan-Red River Fault), DBPF (Dien Bien Phu Fault), GLG F. (Gaoligong Fault), MJ (Mojiang), CN (Changning), ML (Menglian), CL (Chianglai), and CM (Chiangmai). References of paleomagnetic studies: A2013 (Ali et al., 2013), C2012 (Cheng et al., 2012), C2013 (Cheng et al., 2013), C2016 (Chi et al., 2016), HO1991 (Huang & Opdyke, 1991), H1992 (Huang et al., 1992), HO2016 (Huang & Opdyke, 2016), M2019 (Ma et al., 2019), S2012 (Song et al., 2012), S2015 (Song et al., 2015), S2017 (Song et al., 2017), X2014 (Xu et al., 2014), YB1993 (Yang & Besse, 1993), Y2017 (Yan et al., 2017), Y2018 (Yan et al., 2018), Z2015 (Zhao et al., 2015), and Z2019 (Zhao et al., 2019). (b) Geological map and sampling locations in the south Lancangjiang volcanic belt.

Songpan-Ganze-South China and by the Longmu Co-Shuanghu-Changning-Menglian suture to the southwest against the South Qiangtang-Baoshan-Sibumasu (Li et al., 2016; Metcalfe, 2013; Xu et al., 2015).

To better understand the linkages between the Tibetan and Southeast Asia Blocks, we have conducted a paleomagnetic and geochronologic study in the southern Lancangjiang Triassic volcanic belt in western Yunnan of China. The southern Lancangjiang Triassic volcanic belt records volcanic activities during the closure of Paleo-Tethys, a suture zone now located between the Baoshan and Simao Blocks. We synthesize our new data with reported paleomagnetic, geochemical, geochronological, and sedimentary evidences from the Paleo-Tethys realm and provide a comprehensive reconstruction of the paleogeographic configuration of this region.

2. Geologic Setting and Sampling

As the joint of the Tibetan and Southeast Asia realms (Figure 1a), the western Yunnan of China is a key area to investigate their geographic linkage, as well as the evolution of the Paleo-Tethys Ocean, because detailed

19448007, 2019, 24, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019GL085473 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

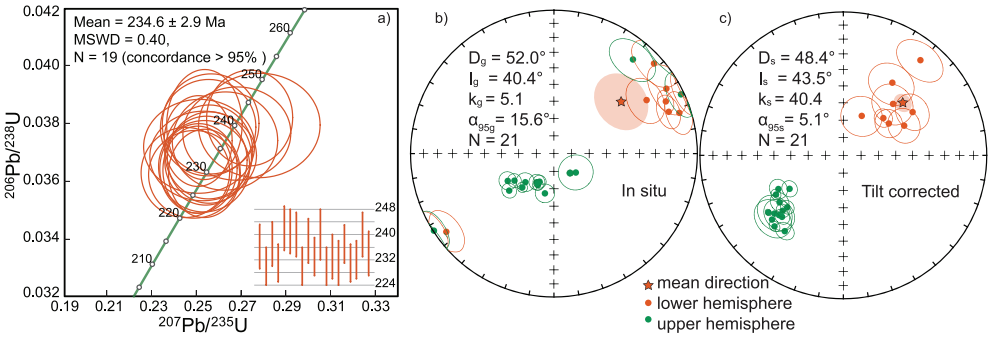


Figure 2. (a) U-Pb dating on zircons showing in the Concordia and Weighted mean diagrams for sample YX7X. Nineteen out of 25 zircons with concordance larger than 95% are used to calculate the weighted average age. Analyzing details are summarized in Table S1 and Figure S2. Equal-area projection of the site means directions of the ChRMs (b) before and (c) after tilt correction. D_g, I_g/D_s, I_s: declination and inclination before and after tilt correction; k and α_{95}: precision parameter and the radius of the cone of 95% confidence for mean-direction; N: number of sites used to calculate. Pmagpy software package are used to plot (Tauxe et al., 2016). Analyzing details are further described in the supporting information.

geological reports are preserved from the Ailaoshan and Lancangjiang zones. The Lancangjiang volcanic belt is at the eastern margin of the Changning-Menglian suture belt, located between the Simao-Indochina Block to the east and the Baoshan-Sibumasu Block to the west (Figure 1). During the subduction of the Paleo-Tethys, the Lincang granitic batholith and the Triassic volcanic rocks formed along the southern Lancangjiang belt (Figure 1b).

The Triassic volcanic sequence is subdivided into the Manghuai Formation (T_2m), Xiaodingxi Formation (T_3x), and Manghuihe Formation (T_3m; Figure 1b; BGMRYP, 1990; Wang et al., 2010). The lower member (T_2m^1) of the Manghuai Formation consists of siltstone, mudstone, and volcano clastic rocks, and the middle and upper members are composed of rhyolites and volcano clastic rocks ($T_2m^{2\text{-}3}$). This formation records syncollisional magmatic eruptions in a continental margin volcanic arc setting (Wang et al., 2010; Wang et al., 2018). With the aim of quantifying the paleo-configuration of the Paleo-Tethys subduction zone in this region, we have collected samples from volcano clastic rocks (Figures S1b–S1e in the supporting information) and rhyolites (Figures S1f–S1h) in the syncollisional Manghuai Formation.

The best outcrops of the Manghuai Formation were found in the Xiaodingxi (eastern Yunxian) and Nuozhadu areas along the Lancangjiang valley (Figure 1b). In the Nuozhadu area, 96 samples at 11 sites were drilled from two rhyolite sections exhibiting clear bedding attitudes (Figures S1f–S1h). As shown in Figures S1f and S1g, beddings are clear to recognize in the rhyolite sections, and in Figure S3a, the microscopic image shows a layer with crystallized quartz and feldspar distributed linearly between noncrystallized glassy materials. The rhythmic layers of quartz and feldspar crystals are recognizable in the outcrops and indicate the attitude of the lava flows during cooling. $^{206}Pb/^{238}U$ age of the rhyolites has been dated as 230±2 Ma (Peng et al., 2013). In the Xiaodingxi (eastern Yunxian) area, samples were collected from 12 sites (131 samples) in a volcano clastic rock section along the valley-road linking Xiaodingxi village and Manghuai town. At least six eruptive cycles were sampled based on field observations (Figures S1b–S1e). Volcanic textures indicate a eutaxitic structure that welded in hot conditions.

3. U-Pb Zircon Dating of Volcanic Rocks

Zircons extracted from volcano clastic rocks are mostly euhedral or subhedral crystals, ranging between 50 and 100 μm in size (Figure S2a). A total of 25 zircons was analyzed, and all U-Pb ages have concordance >90% (Table S2 and Figure S2b in the supporting information). $^{206}Pb/^{238}U$ ages are used in the calculation because all zircon grains are younger than 1.0 Ga. Nineteen of the 25 zircons whose concordance lager than 95% are used to calculate a weighted average age of 234.6±2.9 Ma with mean square weighted deviation of 0.40 (Figure 2a). We thus confidently believe that the Xiaodingxi volcano clastic rocks and Nuozhadu rhyolites sampled in this study are approximately 235–230 Ma in age, which assigns them to the Carnian Stage, the earliest stage of the Late Triassic epoch.

19448007, 2019, 24, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019GL085473 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19448007, 2019, 24, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019GL085473 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

4. Paleomagnetic Analyses

4.1. Demagnetization Results

Following petrographic and rock magnetic analyses (Text S3 and Figures S3 and S4 in the supporting information), ChRMs with dual polarity were isolated from 21 out of the 23 sampling sites, of which 11 sites came from the Nuozhadu rhyolites and 10 sites from the Xiaodingxi volcano clastic rocks (Table S1). For rhyolites specimens, high-temperature components decrease straightforwardly to the origin point following removal of a very low-temperature component by 150°C. More than 80% of the natural remanent magnetizations was only erased between 660°C and 700°C with a step-spacing of 5°C, yielding stable ChRMs carried by hematite (Figures S5a–S5e). In the volcano clastic rock specimens, as illustrated in Figures S5f, S5g, and S5i, following removal of the low-temperature components below 400°C, continuous unblocking above 400°C yields ChRMs recorded by both magnetite and hematite.

4.2. Field Tests and Paleomagnetic Stability

The site mean magnetic direction is calculated to be $D_g/I_g = 52.0°/40.4°$ ($\kappa_g = 5.1$, $\alpha_{95g} = 15.6°$) before and $D_s/I_s = 48.4°/43.5°$ ($\kappa_s = 40.4$, $\alpha_{95s} = 5.1°$) after tilt correction (Figure 2b and Table S1). The data pass McElhinny's (1964) fold test with ks/kg = 7.92 > $F(40,40)_{99\%} = 2.12$ and $F(40,40)_{95\%} = 1.69$ and yield a positive McFadden's (1990) fold test result (in situ $\xi 2 = 19.263$, tilt-corrected $\xi 2 = 2.711$ with $\xi 2$ critical value = 5.207 and 7.300 at 95% and 99% confidence level for N=21). A stepwise tectonic correction (Watson & Enkin, 1993) yields an optimal clustering at 91.4 ± 4.6% unfolding level. Eigen analysis approach suggests that the 95% confident bounds (83–104%) include 100% and thus indicates a pretilt magnetization (Tauxe & Watson, 1994). The new data pass the reversal test (McFadden & Mcelhinny, 1990) with $r = 4.94° < r_{crit.} = 10.1°$. Moreover, it can pass the conglomerate test and paleosecular variation check (Texts S5 and S6 and Figure S6) and is suitable for tectonic reconstruction.

5. Discussions

5.1. Late Triassic Position of the Paleo-Tethys Suture Zone in Western Yunnan

Following the U-Pb dating on zircons, petrographic, rock magnetic, and paleomagnetic analyses, we conclude that the mean pole (46.1°N/176.2°E, K=44.9, A_{95}=4.8°, N=21) derived from the virtual geomagnetic poles of 21 sites in this study is a robust paleomagnetic data with tight age constraint (approximately 235–230 Ma, Carnian Age of Late Triassic). The high-quality data fulfill all the seven Van der Voo's (1990) reliability criteria. The corresponding paleolatitude at the reference site (23.0°N/100.4°E) in the southern Lancangjiang area is 26.0 ± 4.8°N and implies that the paleomagnetic data obtained from the Yiwanshui Formation in Mojiang (Huang & Opdyke, 2016) are very likely an underestimate (approximately 16.8°N) of the Late Triassic paleolatitude of the Simao-Indochina Block. Taking the lithologic property (purplish red or dark purple mudstones and muddy sandstones) and the elongate distribution of the magnetic directions as shown in Huang and Opdyke (2016) into consideration, an empirically based flattening factor $f = 0.6$ (Torsvik et al., 2012; Van der Voo et al., 2015) is utilized to correct the inclination shallowing bias for the data of the Yiwanshui Formation. The corrected inclination (45.3°) and paleolatitude (26.8°N) agree well with the volcanic-derived data in this study (Figure 3a).

In the perspective of Late Triassic paleogeography, the Indochina and North Qiangtang Blocks are interpreted to have been located at 29.0±7.2°N and 31.7±3.0°N (Song et al., 2015; Yan et al., 2017) respectively. Hence, the new dataset in this study supports the Late Triassic paleogeographic reconstructions that have linked Indochina with North Qiangtang (Figure 3b), which was seriously challenged by the data reported by Huang and Opdyke (2016).

5.2. Linkage of the Indochina and North Qiangtang Blocks

The connection between the Qiangtang Block and the Southeast Asian Blocks has been disputed for decades, as summarized in section 1.

Late Paleozoic cold-water faunas, *Glossopteris* floras, as well as Late Carboniferous-Early Permian glacial-marine diamictites provide identical records of Gondwana-affinity linking the proximities of continental blocks (Metcalfe, 2013; Sengor, 1987). The glacial-marine diamictites interbedded with marine clastics were found in the Jilong Formation (P_1j) in the Himalaya, the Laga Formation(P_1l) in the Gangdise, the Zhanjin

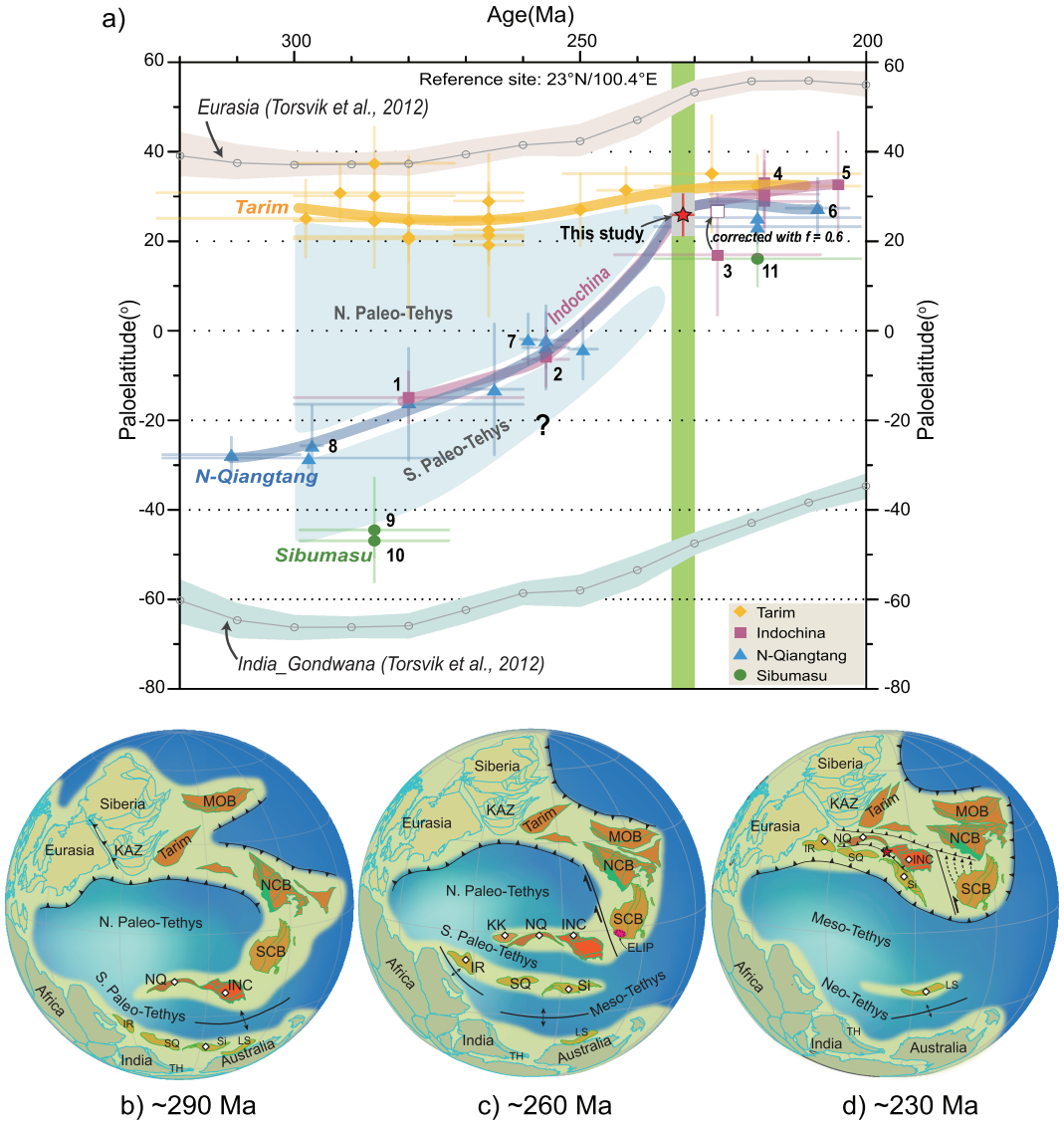


Figure 3. (a) Paleolatitude evolution of the Indochina, North Qiangtang, Sibumasu (Baoshan), Tarim, Eurasia, and Gondwana (India) Blocks. The horizontal and vertical error bars show the uncertainties of ages and paleolatitudes. The paleolatitude errors of Eurasia and India of Gondwana are shown with color strips. Key references: 1. Yan et al., 2018; 2. Chi et al., 2016; 3. Huang & Opdyke, 2016 (the corrected data with $f = 0.6$ is shown with open purple square); 4. Yan et al., 2017; 5. Yan et al., 2019; 6. Song et al., 2015; 7. Ma et al., 2019; 8. Song et al., 2017; 9. Huang & Opdyke, 1991; 10. Xu et al., 2014; 11. Zhao et al., 2015; N. Paleo-Tethys: the north branch of the Paleo-Tethys Ocean; S. Paleo-Tethys: the south branch of the Paleo-Tethys Ocean. All data used are summarized in Table S4. (b–d) Paleogeographic reconstruction of Tethyan realm blocks during the Early Permian to Late Triassic. Paleomagnetic constraints (white diamonds) and the references: (b) NQ: Song et al. (2017), INC: Yan et al. (2018), Si: Huang and Opdyke (1991) & Xu et al. (2014); (c) KK and IR: Muttoni et al. (2009), NQ: Ma et al. (2019), INC: Chi et al. (2016), Si: Zhao et al. (2019), (d) IR: Besse et al. (1998), NQ: Song et al. (2015), INC: Yan et al. (2017), Si: Zhao et al. (2019), LS: Zhou et al. (2016). NCB: North China Block; SCB: South China Block; NQ: North Qiangtang, SQ: South Qiangtang, INC: Indochina, MOB (Mongolia Block), KAZ (Kazakhstan), Si: Sibumasu, IR: Iran, LS: Lhasa, TH: Tethys-Himalaya, KK: Karakoram, ELIP: Emeishan Large Igneous Province.

Formation(P_1z) in the South Qiangtang, and the Dingjiazhai Formation(P_1d) in the Baoshan area and collectively indicate a northward deepening trend of the ocean at the northern margin of Gondwana (Fan et al., 2015; Li et al., 2016). The interpretation of geographic affinity is supported by the analyses of detrital zircons (Gehrels et al., 2011; Li et al., 2016) and paleomagnetic studies (Ali et al., 2013; Huang & Opdyke, 1991; Xu et al., 2014). Early Permian paleomagnetic data have been repeatedly reported from the

19448007, 2019, 24, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019GL085473 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19448007, 2019, 24, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019GL085473 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Woniusi Formation basalts in the Baoshan area and now confirm an approximately 38–42°S paleolatitude (Figures 3a and 3b). Thus, the South Qiangtang-Baoshan Blocks were believed to be located at the northern margin of Gondwana during the Early Permian, though reliable Permian paleomagnetic data have not yet been obtained from the South Qiangtang Block.

In contrast, coeval warm-water sedimentary sequences including coral-reef limestones and fusulinid fossils have been found in the Walongshan ($C_{1\text{-}2}$w), Zharigen (CPz), Changshehu (P_1ch), and Xueyuanhe (P_2x) formations in the North Qiangtang Block, the Longdonghe (C_2l) and Caobashantou(P_1c) formations in the Simao area, and the Saraburi Group (C_3-P_2) in the central Indochina Block (Thailand; Zhong, 1998; Li et al., 2016; Ridd et al., 2011). Although Gehrels et al. (2011) claim that Carboniferous-Permian diamictites were found in northern Qiangtang and similar age distributions of detrital zircons were revealed from both the northern and southern portion of the Qiangtang Block, their sampling sites, located south to the Hongjishan-Xiangtaohu-Guoganjianianshan ophiolite belt of the Longmu Co-Shuanghu suture, are considered to belong to the South Qiangtang Block by other researchers (Fan et al., 2015; Zhu et al., 2013). From the paleomagnetic evidences (Figure 3a), it is apparent that a Permian latitudinal migration of the North Qiangtang Block drifted from approximately 22–24°S to approximately 7.6°S during the Early Permian to Late Permian (Cheng et al., 2013; Ma et al., 2019; Song et al., 2017). This is highly consistent with the Permian drifting history of the Indochina Block whose Early-Middle and Late Permian paleolatitudes are interpreted to have been 21.2±5.7°S (Yan et al., 2018) and approximately 5.3°S (Chi et al., 2016). The tropical position of the North Qiangtang and Indochina Blocks is consistent with the sedimentary and biogeographic evidences as discussed above. Combining their Late Triassic geographic linkage as discussed in section 5.1, the synchronous drifting history during the Early Permian to Late Triassic period suggests a close paleogeographic affinity between the Indochina and North Qiangtang Blocks (Figure 3a).

5.3. Constraints on the Evolution of the Paleo-Tethys Ocean

The Pangea supercontinent was not formed until the Late Carboniferous (~320 Ma) when the Gondwana merged with Laurussia (Torsvik et al., 2012). The oceanic embayment between Gondwana and Laurussia prior to the closure of the Rheic Ocean is termed the Proto-Tethys (Stampfli & Borel, 2002). During the Early Permian, the ocean between Gondwana and Laurussia was the originally defined Paleo-Tethys (Figure 3b). The North Qiangtang-Indochina was located at approximately 22–24°S, which had already drifted from the northern margin of Gondwana since the Early Devonian (Metcalfe, 2013). During the Permian, different branches existed in the Paleo-Tethys Ocean (Figure 3b). The south branch of Paleo-Tethys has also been termed the Longmu Co-Shuanghu-Lancangjiang Ocean by Li et al. (2016), while the north branch between the North Qiangtang-Indochina-South China in the south and the Tarim-Qaidam-North China in the north is termed Mianlue Ocean (Zhao et al., 2018) or Jinshajiang-Ailaoshan Ocean (Jian et al., 2009a, 2009b; Wang et al., 2000).

During the Late Permian, the North Qiangtang-Indochina are interpreted to have been located in equatorial areas (Figure 3c; Huang et al., 1992; Chi et al., 2016; Ma et al., 2019), as well as the Karakoram (Muttoni et al., 2009). Their breakup with the South China Block is interpreted to be trigged by the eruption of the Emeishan Large Igneous Province (Chung et al., 1998). At the same time, we know little about the position of the South Qiangtang and Sibumasu (Zhao et al., 2019), though the Iranian Blocks are interpreted to have drifted from Gondwana by the end of the Permian (Muttoni et al., 2009).

By comparing the paleolatitudes of the North Qiangtang against that of the Tarim-North China, Song et al. (2015) proposed that the north branch of Paleo-Tethys had closed by the Late Triassic, strictly, the late Late Triassic (213–204 Ma). However, the new paleolatitude estimate (26.0±4.8°N) derived from the southern Lancangjiang volcanic belt in this study indicates that the North Qiangtang-Indochina had drifted further north to approximately 26°N by the early Late Triassic (235–230 Ma). Therefore, we suggest that the north branch had probably closed no later than the early Late Triassic (235–230 Ma; Figures 3a and 3d). This conclusion is consistent with much geologic evidence (Metcalfe, 2013; Zhao et al., 2018). Although the closure of the south branch of the Paleo-Tethys has not been derived from paleomagnetic evidence, the record of oceanic evolution-related rocks including ophiolite, radiolarian chert, high-pressure eclogite, and blueschist indicates that the closure of the south branch of Paleo-Tethys very likely occurred during the late Middle Triassic to early Late Triassic (Figure 3d; Zhong, 1998; Metcalfe, 2013; Li et al., 2016).

6. Conclusions

We report a robust early Late Triassic paleomagnetic pole with tight age constraint from volcanic rocks in the southern Lancangjiang volcanic belt. The ChRMs pass fold, reversal, and especially conglomerate tests and are very likely primary thermal remanent magnetization records. The new pole (46.1°N/176.2°E, K=44.9, A_{95}=4.8, N=21) corresponds a paleolatitude of 26.0±4.8°N at the reference site (23.0°N/100.4°E) in the southern Lancangjiang area. We suggest a close paleogeographic affinity for Indochina and North Qiangtang Blocks, at least during the Early Permian to Late Triassic interval, and propose that the north branch of the Paleo-Tethys Ocean closed no later than the early Late Triassic.

Acknowledgments

We thank Zhiyu Yi, John D. A. Piper, and Yuejun Wang for their constructive comments and Yulin Han, Xiaodong Tan, Huafeng Qin, Wenxing Hao, Haoyang Zhou, Ying Liu, and Xiaoming Sun for their helpful suggestions in the laboratory analyses. Monika Korte, Yo-ichiro Otofuji, and another anonymous reviewer are also acknowledged for their comments and peer editing. Jianjun Li is also appreciated for field assistance. This work is financially supported by the NSFC (Natural Science Foundation of China) projects (41802237), Guangdong Province Introduced Innovative R&D Team of Geological Processes and Natural Disasters around the South China Sea (2016ZT06N331), and China Postdoctoral Science Foundation (2017M622855 and 2019T120766). Readers can access our data in the supporting information or at https://earthref.org/MagIC/16681/a84486f1-e555-42a4-85b9-9507582e5641.

References

Ali, J. R., Cheung, H. M., Aitchison, J. C., & Sun, Y. (2013). Palaeomagnetic re-investigation of Early Permian rift basalts from the Baoshan Block, SW China: Constraints on the site-of-origin of the Gondwana-derived eastern Cimmerian terranes. *Geophysical Journal International*, *193*, 650–663. https://doi.org/10.1093/gji/ggt012

Besse, J., Torcq, F., Gallet, Y., Ricou, L. E., Krystyn, L., & Saidi, A. (1998). Late Permian to Late Triassic palaeomagnetic data from Iran: Constraints on the migration of the Iranian Block through the Tethyan Ocean and initial destruction of Pangaea. *Geophysical Journal International*, *135*, 77–92. https://doi.org/10.1046/j.1365-246X.1998.00603.x

BGMRYP (Bureau of Geology and Mineral Resources of Yunnan Province) (1990). *Regional Geology of Yunnan Province*, (p. 727). Beijing: Geological Publishing House. (In Chinese)

Cheng, X., Wu, H. N., Diao, Z. B., Wang, H. J., Ma, L., Zhang, X. D., et al. (2013). Paleomagnetic data from the Late Carboniferous-Late Permian rocks in eastern Tibet and their implications for tectonic evolution of the northern Qiangtang-Qamdo Block. *Science China Earth Sciences*, *56*(7), 1209–1220. https://doi.org/10.1007/s11430-012-4558-1

Cheng, X., Wu, H. N., Guo, Q., Hou, B. N., Xia, L. Y., Wang, H. J., et al. (2012). Paleomagnetic results of Late Paleozoic rocks from northern Qiangtang Block in Qinghai-Tibet Plateau, China. *Science China Earth Sciences*, *55*(1), 67–75. https://doi.org/10.1007/s11430-011-4287-x

Chi, C. T., Geissman, J. W., Quy, H. V., Anh, T. V., & Dung, N. T. P. (2016). Paleomagnetism of Upper Permian basaltic rocks of Cam Thuy formation from Thuan Chau locality, Son La, Northwest Vietnam. *Vietnam Journal of Earth Sciences*, *37*, 289–298. https://doi.org/10.15625/0866-7187/37/4/8057

Chung, S. L., Jahn, B. M., Genyao, W., Lo, C. H., & Bolin, C. (1998). The Emeishan flood basalt in SW China: A mantle plume initiation model and its connection with continental breakup and mass extinction at the Permian-Triassic boundary. In M. F. J. Flower, S. L. Chung, C. H. Lo, & T. Y. Lee (Eds.), *Mantle dynamics and plate interactions in East Asia*, (Vol. 27, pp. 47–58). Washington DC: AGU. https://doi.org/10.1029/GD027p0047

Fan, J. J., Li, C., Wang, M., Xie, C.-M., & Xu, W. J. G. R. (2015). Features, provenance, and tectonic significance of Carboniferous–Permian glacial marine diamictites in the Southern Qiangtang–Baoshan Block, Tibetan Plateau. *Gondwana Research*, *28*, 1530–1542. https://doi.org/10.1016/j.gr.2014.10.015

Gehrels, G., Kapp, P., Decelles, P., Pullen, A., Blakey, R., Weislogel, A., et al. (2011). Detrital zircon geochronology of pre-Tertiary strata in the Tibetan-Himalayan orogen. *Tectonics*, *30*, TC5016. https://doi.org/10.1029/2011TC002868

Haag, M., & Heller, F. (1991). Late Permian to Early Triassic magnetostratigraphy. *Earth and Planetary Science Letters*, *107*, 42–54. https://doi.org/10.1016/0012-821X(91)90042-G

Huang, K., & Opdyke, N. D. (1991). Paleomagnetic results from the Upper Carboniferous of the Shan-Thai-Malay Block of western Yunnan, China. *Tectonophysics*, *192*, 333–344. https://doi.org/10.1016/0040-1951(91)90107-4

Huang, K., & Opdyke, N. D. (2016). Paleomagnetism of the Upper Triassic rocks from south of the Ailaoshan Suture and the timing of the amalgamation between the South China and the Indochina Blocks. *Journal of Asian Earth Sciences*, *119*, 118–127. https://doi.org/10.1016/j.jseaes.2015.12.005

Huang, K., Opdyke, N. D., Peng, X., & Li, J. (1992). Paleomagnetic results from the Upper Permian of the eastern Qiangtang Terrane of Tibet and their tectonic implications. *Earth and Planetary Science Letters*, *111*, 1–10. https://doi.org/10.1016/0012-821X(92)90164-Q

Jian, P., Liu, D., Kröner, A., Zhang, Q., Wang, Y., Sun, X., & Zhang, W. (2009a). Devonian to Permian plate tectonic cycle of the Paleo-Tethys Orogen in southwest China (I): Geochemistry of ophiolites, arc/back-arc assemblages and within-plate igneous rocks. *Lithos*, *113*, 748–766. https://doi.org/10.1016/j.lithos.2009.04.004

Jian, P., Liu, D., Kröner, A., Zhang, Q., Wang, Y., Sun, X., & Zhang, W. (2009b). Devonian to Permian plate tectonic cycle of the Paleo-Tethys Orogen in southwest China (II): Insights from zircon ages of ophiolites, arc/back-arc assemblages and within-plate igneous rocks and generation of the Emeishan CFB province. *Lithos*, *113*, 767–784. https://doi.org/10.1016/j.lithos.2009.04.006

Li, C., Xie, C. M., Wang, M., Wu, Y. W., Hu, P. Y., Zhang, X. Z., et al. (2016). *Geology of Qiangtang*. Beijing: Geologic Press. (In Chinese)

Ma, Y., Wang, Q., Wang, J., Yang, T., Tan, X., Dan, W., et al. (2019). Paleomagnetic constraints on the origin and drift history of the North Qiangtang Terrane in the Late Paleozoic. *Geophysical Research Letters*, *46*, 689–697. https://doi.org/10.1029/2018GL080964

McElhinny, M. W. (1964). Statistical significance of the fold test in palaeomagnetism. *Geophysical Journal International*, *8*, 338–340. https://doi.org/10.1111/j.1365-246X.1964.tb06300.x

McFadden, P. L. (1990). A new fold test for palaeomagnetic studies. *Geophysical Journal International*, *103*, 163–169. https://doi.org/10.1111/j.1365-246X.1990.tb01761.x

McFadden, P. L., & Mcelhinny, M. W. (1990). Classification of the reversal test in paleomagnetism. *Geophysical Journal International*, *103*, 725–729. https://doi.org/10.1111/j.1365-246X.1990.tb05683.x

Metcalfe, I. (2013). Gondwana dispersion and Asian accretion: Tectonic and palaeogeographic evolution of eastern Tethys. *Journal of Asian Earth Sciences*, *66*, 1–33. https://doi.org/10.1016/j.jseaes.2012.12.020

Muttoni, G., Gaetani, M., Kent, D. V., Sciunnach, D., Angiolini, L., Berra, F., et al. (2009). Opening of the Neo-Tethys Ocean and the Pangea B to Pangea A transformation during the Permian. *GeoArabia*, *14*, 17–48.

Peng, T., Wilde, S. A., Wang, Y., Fan, W., & Peng, B. (2013). Mid-Triassic felsic igneous rocks from the southern Lancangjiang Zone, SW China: Petrogenesis and implications for the evolution of Paleo-Tethys. *Lithos*, *168–169*, 15–32. https://doi.org/10.1016/j.lithos.2013.01.015

Ridd, M. F., Barber, A. J., & Crow, M. J. (2011). *The geology of Thailand*. London: Geological Society.

19448007, 2019, 24, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019GL085473 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Royden, L. H., Burchfiel, B. C., & Rd, V. D. H. (2008). The geological evolution of the Tibetan Plateau. *Science*, *321*, 1054–1058. https://doi.org/10.1126/science.1155371

Sengor, A. (1987). Tectonics of the Tethysides: Orogenic collage development in a collisional setting. *Annual Review of Earth and Planetary Sciences*, *15*, 213–244.

Song, C. Y., Wang, J., Fu, X. G., Feng, X. L., Chen, M., & He, L. (2012). Late triassic paleomagnetic data from the Qiangtang terrane of Tibetan Plateau and their tectonic significances. *Journal of Jilin University*, *42*, 526–535. (In Chinese with English abstract)

Song, P., Ding, L., Li, Z., Lippert, P. C., Yang, T., Zhao, X., et al. (2015). Late Triassic paleolatitude of the Qiangtang Block: Implications for the closure of the Paleo-Tethys Ocean. *Earth and Planetary Science Letters*, *424*, 69–83. https://doi.org/10.1016/j.epsl.2015.05.020

Song, P., Ding, L., Li, Z., Lippert, P. C., & Yue, Y. (2017). An early bird from Gondwana: Paleomagnetism of Lower Permian lavas from northern Qiangtang (Tibet) and the geography of the Paleo-Tethys. *Earth and Planetary Science Letters*, *475*, 119–133. https://doi.org/10.1016/j.epsl.2017.07.023

Stampfli, G., & Borel, G. (2002). A plate tectonic model for the Paleozoic and Mesozoic constrained by dynamic plate boundaries and restored synthetic oceanic isochrons. *Earth and Planetary Science Letters*, *196*, 17–33. https://doi.org/10.1016/S0012-821X(01)00588-X

Tauxe, L., Shaar, R., Jonestrask, L., Swanson-Hysell, N., Minnett, R., Koppers, A., et al. (2016). PmagPy: Software package for paleomagnetic data analysis and a bridge to the Magnetics Information Consortium (MagIC) Database. *Geochemistry, Geophysics, Geosystems*, *17*, 2450–2463. https://doi.org/10.1002/2016GC006307

Tauxe, L., & Watson, G. S. (1994). The fold test: An Eigen analysis approach. *Earth and Planetary Science Letters*, *122*, 331–341. https://doi.org/10.1016/0012-821X(94)90006-X

Torsvik, T. H., Van der Voo, R., Preeden, U., Mac Niocaill, C., Steinberger, B., Doubrovine, P. V., et al. (2012). Phanerozoic Polar Wander, palaeogeography and dynamics. *Earth Science Reviews*, *114*, 325–368. https://doi.org/10.1016/j.earscirev.2012.06.007

Van der Voo, R. (1990). The reliability of paleomagnetic data. *Tectonophysics*, *184*, 1–9. https://doi.org/10.1016/0040-1951(90)90116-P

Van Der Voo, R. (2005). *Paleomagnetism of the Atlantic, Tethys and Iapetus oceans*.Cambridge, UK: Cambridge University Press.

Van der Voo, R., van Hinsbergen, D. J. J., Domeier, M., Spakman, W., & Torsvik, T. H. (2015). Latest Jurassic–earliest Cretaceous closure of the Mongol-Okhotsk Ocean: A paleomagnetic and seismological-tomographic analysis. *Geological Society of America Special Papers*, *513*, 589–606. https://doi.org/10.1130/2015.2513(19)

van Hinsbergen, D. J. J., Kapp, P., Dupont-Nivet, G., Lippert, P. C., DeCelles, P. G., & Torsvik, T. H. (2011). Restoration of Cenozoic deformation in Asia and the size of Greater India. *Tectonics*, *30*, TC5003. https://doi.org/10.1029/2011TC002908

Wang, X., Metcalfe, I., Jian, P., He, L., & Wang, C. (2000). The Jinshajiang–Ailaoshan suture zone, China: Tectonostratigraphy, age and evolution. *Journal of Asian Earth Sciences*, *18*, 675–690. https://doi.org/10.1016/S1367-9120(00)00039-0

Wang, Y., Qian, X., Cawood, P. A., Liu, H., Feng, Q., Zhao, G., et al. (2018). Closure of the East Paleotethyan Ocean and amalgamation of the Eastern Cimmerian and Southeast Asia continental fragments. *Earth Science Reviews*, *186*, 195–230. https://doi.org/10.1016/j.earscirev.2017.09.013

Wang, Y., Zhang, A., Fan, W., Peng, T., Zhang, F., Zhang, Y., & Bi, X. (2010). Petrogenesis of late Triassic post-collisional basaltic rocks of the Lancangjiang tectonic zone, southwest China, and tectonic implications for the evolution of the eastern Paleotethys: Geochronological and geochemical constraints. *Lithos*, *120*, 529–546. https://doi.org/10.1016/j.lithos.2010.09.012

Watson, G. S., & Enkin, R. J. (1993). The fold test in paleomagnetism as a parameter estimation problem. *Geophysical Research Letters*, *20*, 2135–2137. https://doi.org/10.1029/93GL01901

Xiao, W. J., Wendley, B. F., Liu, D. Y., Jian, P., Liu, C. Z., Yian, C., & Sun, M. (2005). Accretionary tectonics of the Western Kunlun Orogen, China: A Paleozoic–Early Mesozoic, long-lived active continental margin with implications for the growth of Southern Eurasia. *Journal of Geology*, *113*, 687–705.

Xu, Y., Yang, Z., Tong, Y. B., Wang, H., Gao, L., & An, C. (2014). Further paleomagnetic results for lower Permian basalts of the Baoshan Terrane, southwestern China, and paleogeographic implications. *Journal of Asian Earth Sciences*, *104*, 318–326. https://doi.org/10.1016/j.jseaes.2014.10.029

Xu, Z., Dilek, Y., Cao, H., Yang, J., Robinson, P., Ma, C., et al. (2015). Paleo-Tethyan evolution of Tibet as recorded in the East Cimmerides and West Cathaysides. *Journal of Asian Earth Sciences*, *105*, 320–337. https://doi.org/10.1016/j.jseaes.2015.01.021

Yan, Y., Huang, B., Zhang, D., Charusiri, P., & Veeravinantanakul, A. (2018). Paleomagnetic Study on the Permian rocks of the Indochina Block and Its Implications for paleogeographic configuration and northward drifting of Cathaysialand in the Paleo-Tethys. *Journal of Geophysical Research: Solid Earth*, *123*, 4523–4538. https://doi.org/10.1029/2018JB015511

Yan, Y., Huang, B., Zhao, J., Zhang, D., Liu, X., Charusiri, P., & Veeravinantanakul, A. (2017). Large southward motion and clockwise rotation of Indochina throughout the Mesozoic: Paleomagnetic and detrital zircon U–Pb geochronological constraints. *Earth and Planetary Science Letters*, *459*, 264–278. https://doi.org/10.1016/j.epsl.2016.11.035

Yan, Y., Zhao, Q., Zhang, D., Charusiri, P., Huang, B., & Zhang, P. (2019). Palaeomagnetism of Late Triassic volcanic rocks from the western margin of Khorat Basin, Thailand and its implication for ambiguous inclination shallowing in Mesozoic sediments of Indochina. *Geophysical Journal International*, *219*, 897–910. https://doi.org/10.1093/gji/ggz331

Yang, Z., & Besse, J. (1993). Paleomagnetic study of Permian and Mesozoic sedimentary rocks from Northern Thailand supports the extrusion model for Indochina. *Earth and Planetary Science Letters*, *117*, 525–552. https://doi.org/10.1016/0012-821X(93)90101-E

Yin, A., & Harrison, T. M. (2000). Geologic evolution of the Himalayan-Tibetan orogen. *Annual Review of Earth and Planetary Sciences*, *28*, 211–280. https://doi.org/10.1146/annurev.earth.28.1.211

Zhao, G., Wang, Y., Huang, B., Dong, Y., Li, S., Zhang, G., & Yu, S. (2018). Geological reconstructions of the East Asian Blocks: From the breakup of Rodinia to the assembly of Pangea. *Earth Science Reviews*, *186*, 262–286. https://doi.org/10.1016/j.earscirev.2018.10.003

Zhao, J., Huang, B., Yan, Y., Bai, Q., Dong, Y., Win, Z., et al., 2019. A palaeomagnetic study of the Middle Permian and Middle Triassic limestones from Shan State, Myanmar: Implications for collision of the Sibumasu Terrane and Indochina Terrane. Geological Journal. 1-16. https://doi.org/10.1002/gj.3482

Zhao, J., Huang, B., Yan, Y., & Zhang, D. (2015). Late Triassic paleomagnetic result from the Baoshan Terrane, West Yunnan of China: Implication for orientation of the East Paleotethys suture zone and timing of the Sibumasu-Indochina collision. *Journal of Asian Earth Sciences*, *111*, 350–364. https://doi.org/10.1016/j.jseaes.2015.06.033

Zhong, D. (1998). *Paleotethyan orogenic belts in Yunnan and Western Sichuan*, (pp. 1–131). Beijing: Science Press. (In Chinese)

Zhou, Y., Cheng, X., Yu, L., Yang, X., Su, H., Peng, X., et al. (2016). Paleomagnetic study on the Triassic rocks from the Lhasa Terrane, Tibet, and its paleogeographic implications. *Journal of Asian Earth Sciences*, *121*, 108–119. https://doi.org/10.1016/j.jseaes.2016.02.006

Zhu, D. C., Zhao, Z. D., Niu, Y., Dilek, Y., Hou, Z. Q., & Mo, X. X. (2013). The origin and pre-Cenozoic evolution of the Tibetan Plateau. *Gondwana Research*, *23*, 1429–1454. https://doi.org/10.1016/j.gr.2012.02.00

19448007, 2019, 24, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019GL085473 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

References From the Supporting Information

Deenen, M. H. L., Langereis, C. G., van Hinsbergen, D. J. J., & Biggin, A. J. (2011). Geomagnetic secular variation and the statistics of palaeomagnetic directions. *Geophysical Journal International*, *186*, 509–520. https://doi.org/10.1111/j.1365-246X.2011.05050.x Fisher, R. (1953). Dispersion on a sphere. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering*, *217*, 295–305. https://doi.org/10.1098/rspa.1953.0064

Heslop, D., Dekkers, M. J., Kruiver, P. P., & van Oorschot, I. H. M. (2002). Analysis of isothermal remanent magnetization acquisition curves using the Expectation-Maximization algorithm. *Geophysical Journal International*, *148*, 58–64. https://doi.org/10.1046/j.0956-540x.2001.01558.x

Jackson, A., Jonkers, A. R. T., & Walker, M. R. (2000). Four centuries of geomagnetic secular variation from historical records. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering*, *358*(1768), 957–990. https://doi.org/10.1098/rsta.2000.0569

Johnson, C. L., & Constable, C. G. (1995). The time-averaged geomagnetic field as recorded by lava flows over the past 5 Myr. *Geophysical Journal International*, *122*, 489–519. https://doi.org/10.1111/j.1365-246X.1995.tb07010.x

Kirschvink, J. L. (1980). The least-squares line and plane and the analysis of paleomagnetic data. *Journal of Geophysical Research*, *62*, 699–718. https://doi.org/10.1111/j.1365-246X.1980.tb02601.x

Ludwig, K. (2008). Isoplot version 4.15: a geochronological toolkit for microsoft Excel. Berkeley Geochronology Cent. *Special Publication*, *4*, 247–270.

O'Reilly, W. (1984). *Rock and Mineral Magnetism*. Glasgow: Blackie.

Tauxe, L. (2010). *Essentials of paleomagnetism*, (p. 178). Berkeley, Los Angeles, London: University of California Press.

Watson, G. S. (1983). Large sample theory of the Langevin distribution. *Journal of Statistical Planning and Inference*, *8*, 245–256. https://doi.org/10.1016/0378-3758(83)90043-5

19448007, 2019, 24, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019GL085473 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License